TOGETHER APP & FARMING SYSTEM

SAVE THE PLANET

BE THE CHANGE

PROBLEM

Agriculture is extremely vulnerable to climate change, as it poses a threat to food security, particularly to agriculture, with yields directly affected by plant diseases which cost the global economy billions and create a threat to food security everywhere.

Lack of awareness, in addition to the wrong belief that proactive action will not make the difference in combating climate change and that the solutions to it are too difficult. But this is untrue, as various solutions exist.

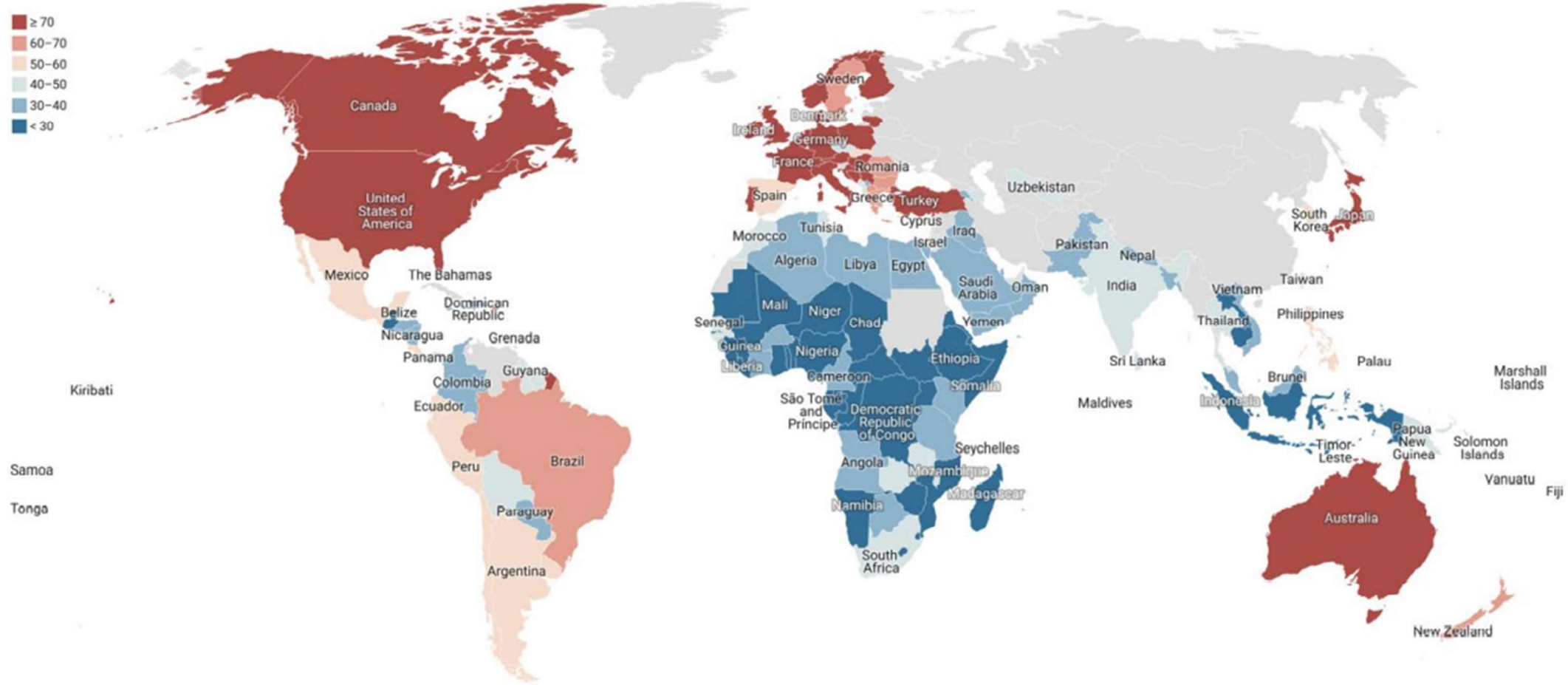
FAO Study : Crop Pests and diseases destroy up to 40 % of global crops and cost $220 billion of losses

Climate change fans spread of pests and threatens plants and crops, new FAO study

Pests destroy up to 40 percent of global crops and cost $220 billion of losses

Knowledge about climate change

% who know 'a moderate amount' or 'a lot' about climate change



Source: Yale Program on Climate Change Communication / Data for Good at Meta • Created with Datawrapper

SOLUTION

Together, a mobile application and smart farming system help mitigate climate change by providing smart agricultural solutions to maximize productivity, in addition to building communities that are more aware of climate change and providing a tool to facilitate planting.

SMART AGRICULTURE SYSTEM

EMPOWERING APP

1

SMART AGRICULTURE SYSTEM

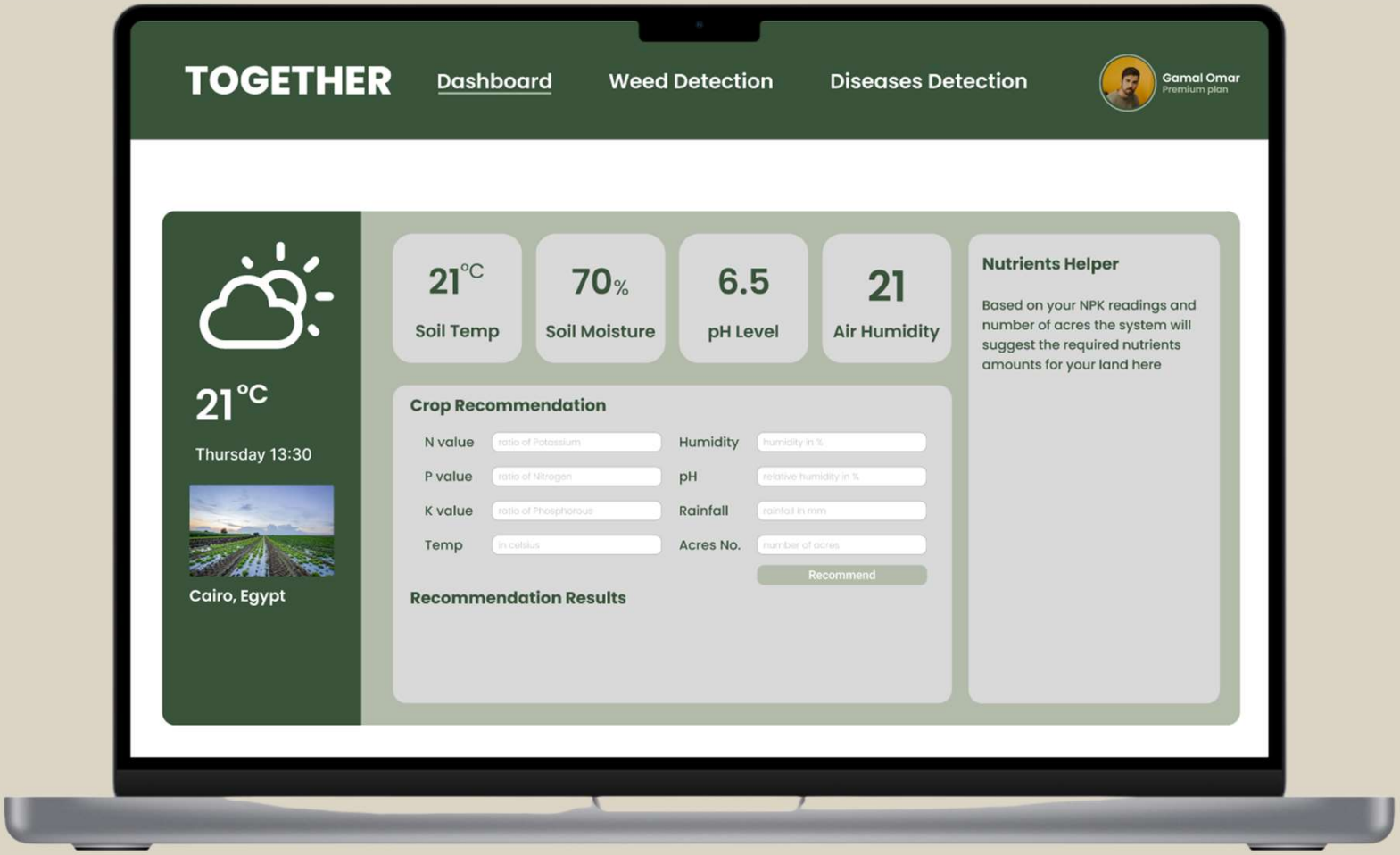
SMART AGRICULTURE SYSTEM

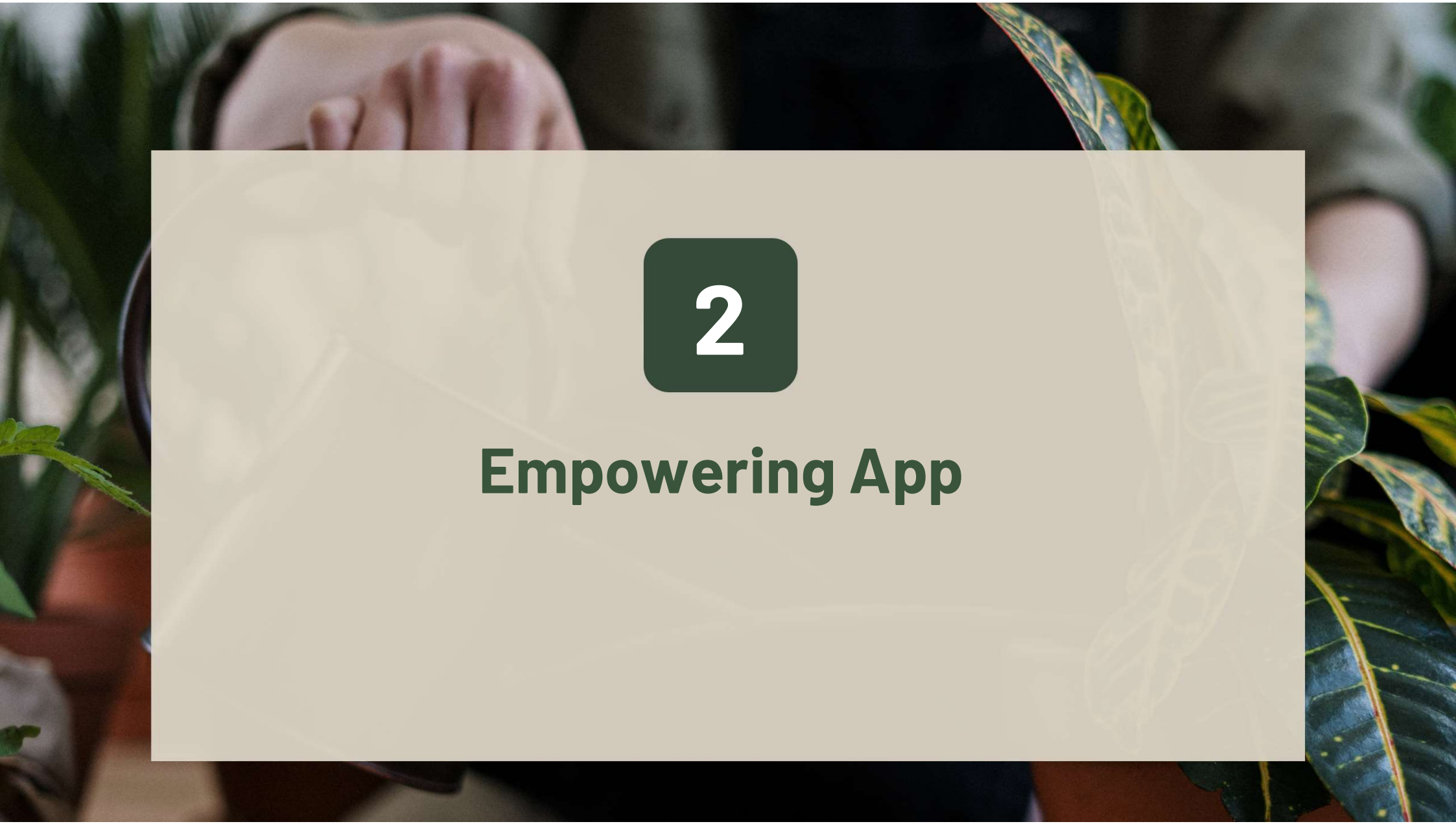
Through web-based system, we are providing 3 services :

INTELLIGENT WEED DETECTION SYSTEM

Intelligent disease detection system

CROP RECOMMENDATION AND SUPPORT SYSTEM


TOGETHER
Dashboard
Weed Detection
Diseases Detection
Gamal Omar
Premium plan
21°C
Thursday 13:30
Cairo, Egypt
21°C
Soil Temp
70%
Soil Moisture
6.5
pH Level
21
Air Humidity
Crop Recommendation
N value
P value
K value
Temp
Humidity
pH
Rainfall
Acres No.
Recommend
Recommendation Results
Nutrients Helper
Based on your NPK readings and number of acres the system will suggest the required nutrients amounts for your land here



2

Empowering App

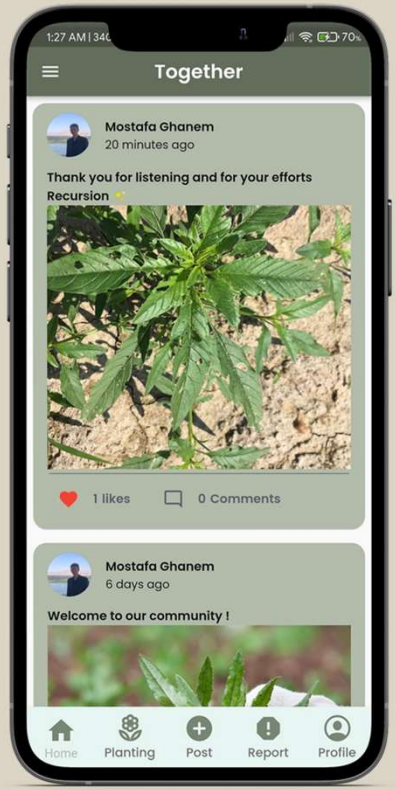
Empowering App

Together app provides a Community, Planting smart guide & Pollution report

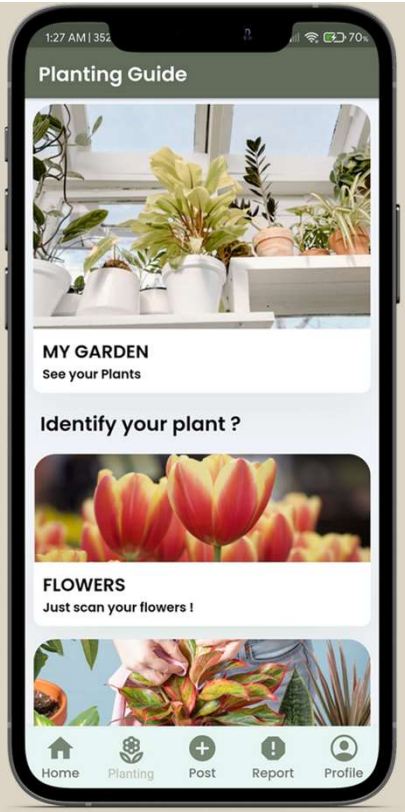
PLANTING SMART GUIDE

POLLUTION REPORTING FEATURE

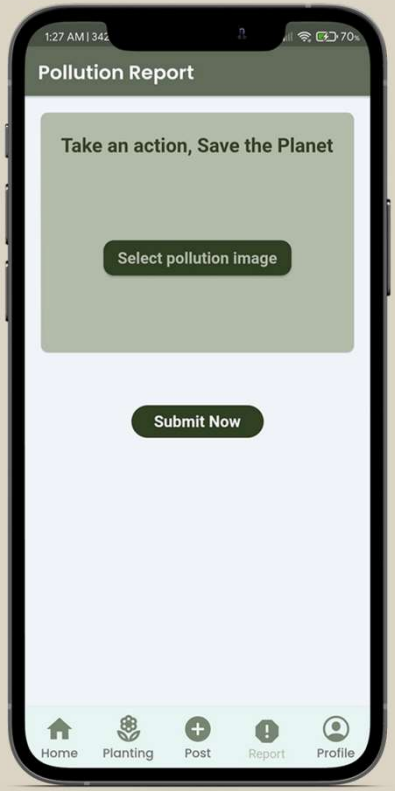
SOCIAL PLATFORM


Together
Mostafa Ghanem
20 minutes ago
Thank you for listening and for your efforts
Recursion
1 likes
0 Comments
Mostafa Ghanem
6 days ago
Welcome to our community !
Home
Planting
Post
Report
Profile

Ahmed Radwan
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Pharetra aliquam, congue habitasse tortor.
2 hrs
Ola Gamal
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Pharetra aliquam, congue habitasse tortor.
30 mins
Manar Khurshid
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Pharetra aliquam, congue habitasse tortor.
20 mins
Real Omar
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Pharetra aliquam, congue habitasse tortor.
Just Now
Write a comment...

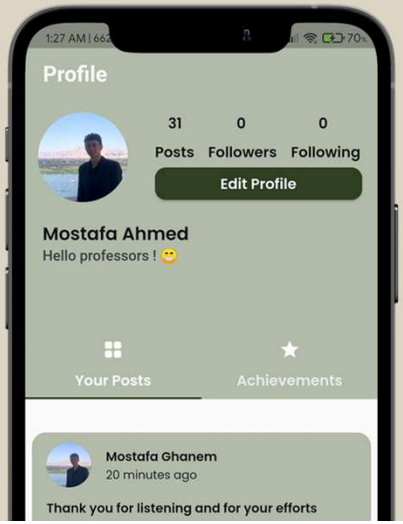

Planting Guide
MY GARDEN
See your Plants
Identify your plant ?
FLOWERS
Just scan your flowers !
Home
Planting
Post
Report
Profile

Mostafa Ghanem
What's happening?
Home
Planting
Post
Report
Profile


Pollution Report
Take an action, Save the Planet
Select pollution image
Submit Now
Home
Planting
Post
Report
Profile

Plant Information
Pterocarya Stenoptera
About
Common Name
Chinese wingnut
Chinese wingnut (Pterocarya stenoptera) is a deciduous tree in the walnut family that will grow from 15 to 21 m tall. Native to China it makes a beautiful shade tree. It blooms from spring to summer with light green pendulous flowers followed by winged seeds that give this tree its

My Garden
Flowering Plants
Sunflower
Leafy Plants

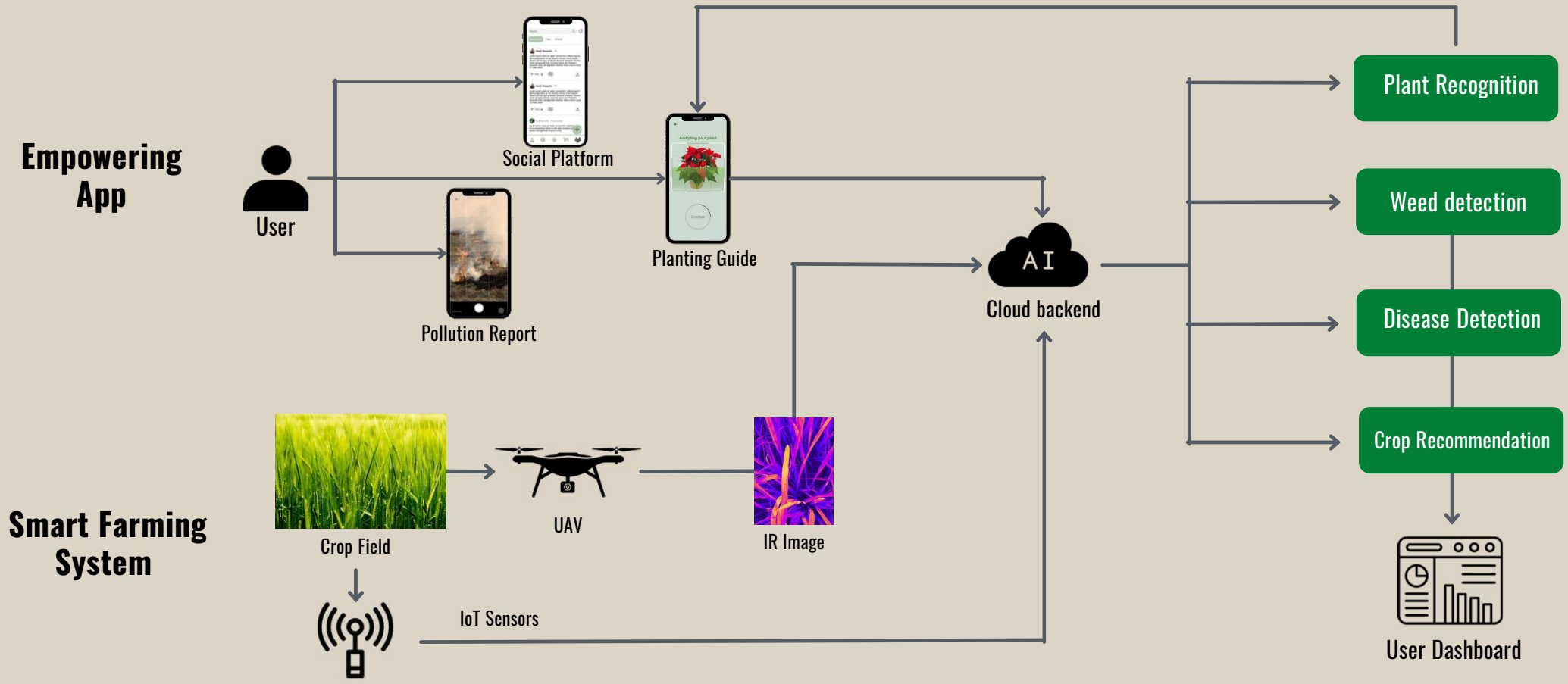

Profile
31
Posts
0
Followers
0
Following
Edit Profile
Mostafa Ahmed
Hello professors !
Your Posts
Achievements
Mostafa Ghanem
20 minutes ago
Thank you for listening and for your efforts

VALUE

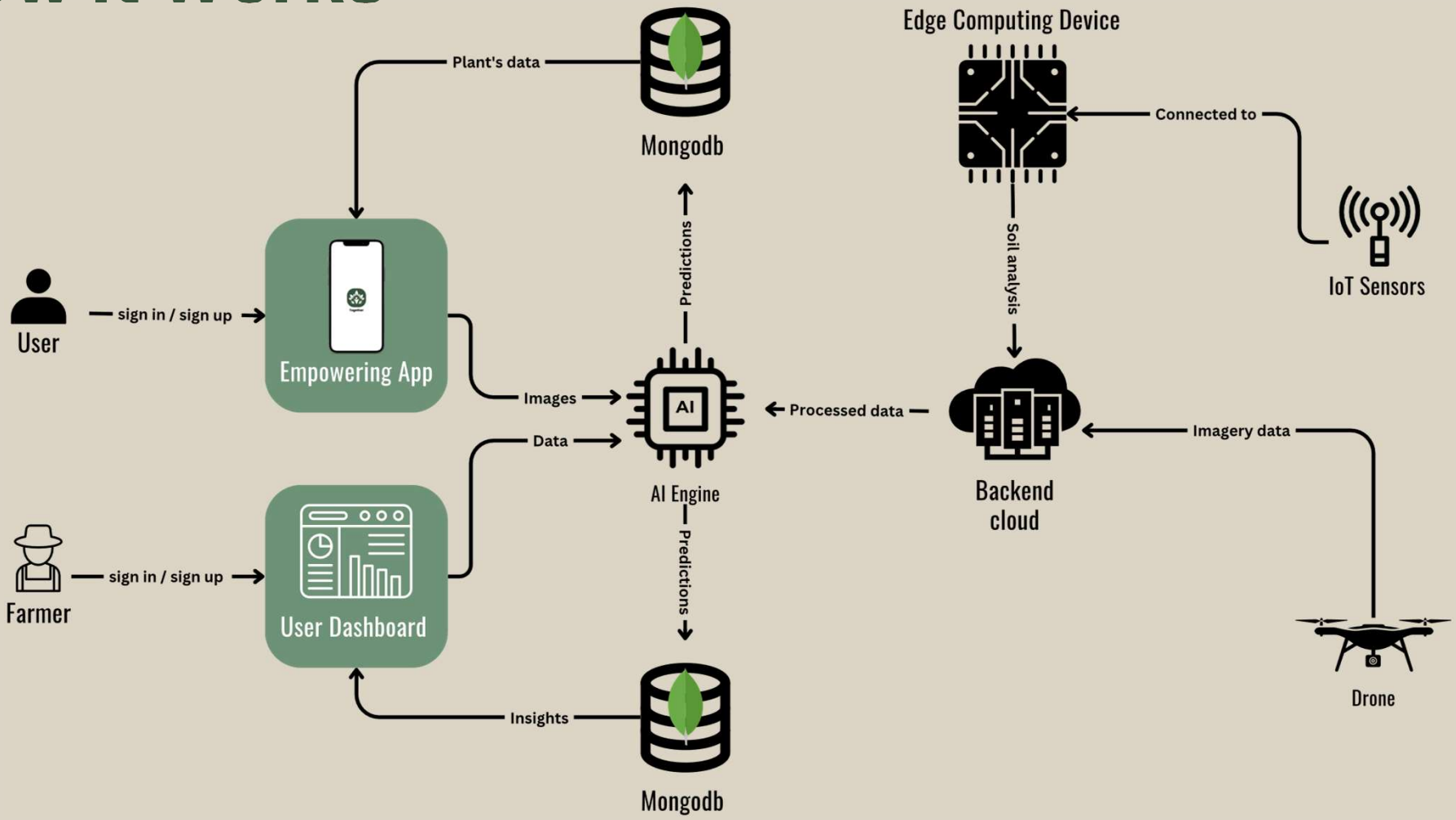
Increase crop yield production and utilize resources usage such as water, fertilizers, and pesticides using AI-powered smart farming techniques.

Increase the community awareness about the climate change and how to take actions by providing tools for them like pollution report and smart planting guide

High-level Architecture



How It Works


Edge Computing Device
Plant's data
Mongodb
Connected to
IoT Sensors
Predictions
Soil analysis
User
sign in / sign up
Empowering App
Images
AI
Processed data
Data
Imagery data
AI Engine
Backend cloud
Farmer
sign in / sign up
User Dashboard
Predictions
Drone
Insights
Mongodb

TECHNOLOGY

DEEP LEARNING MODELS

For Plant Recognition, Disease Detection, Weed Detection and Crop Recommendation.

SMART FARMING SYSTEM WEBSITE

With dashboard and **3** services, a website has been developed to using **ReactJS** and **Flask** to deliver the farming services

MOBILE APP

Using **Flutter** framework, we have developed a mobile app with **Flask**, **PyTorch** and **TensorFlow** for the AI backend and the social platform and database is working with the aid of **Mongo DB** & **Firebase,** we have also deployed our backend to **AWS cloud.**

DATASETS

Crop Recommendation Dataset

Maximize agricultural yield by recommending appropriate crops

k kaggle.com

| Flowers

Can you identify all 16 flower species?

k kaggle.com

Leaf Classification

Can you see the random forest for the leaves?

k kaggle.com

CottonWeedID15

Multiclass Identification of Common Weed Species in Cotton Production Systems

k kaggle.com

V2 Plant Seedlings Dataset

Images of crop and weed seedlings at different growth stages.

k kaggle.com

Paddy Doctor: Paddy Disease Classification

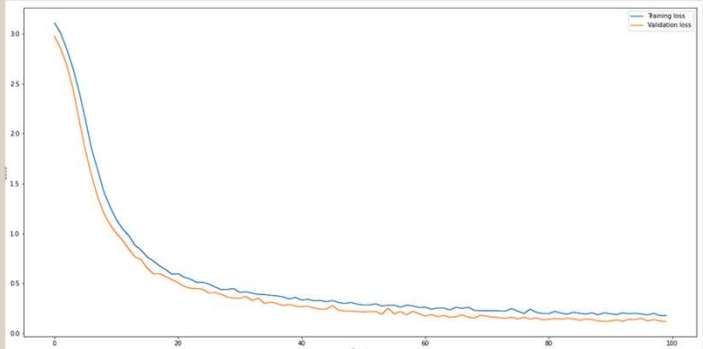
Identify the type of disease present in paddy leaf images

k kaggle.com

Results

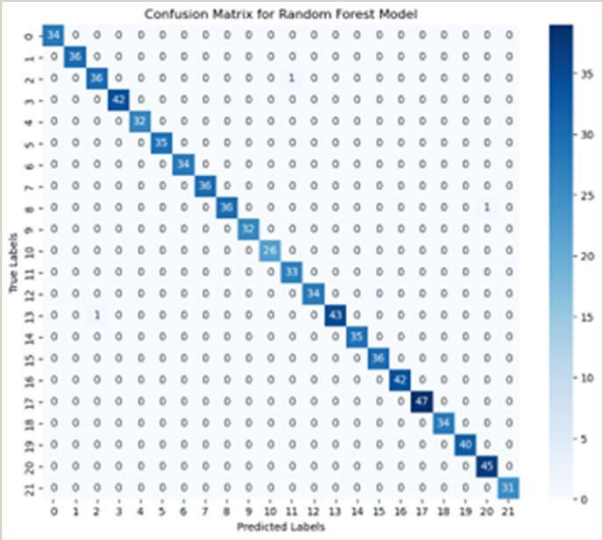
Crop Recommendation

CNN+LSTM Deep Learning Model

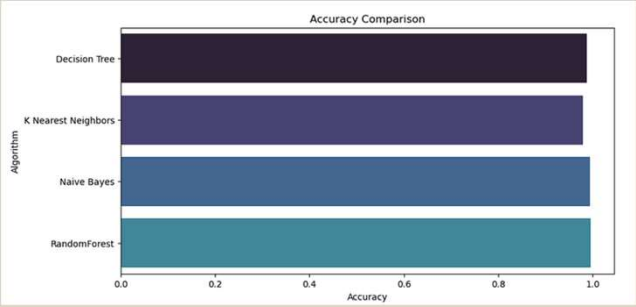



```
14/14 [==============================] - 0s 3ms/step - loss: 0.1082 - accuracy: 0.9659
[0.10821560770273209, 0.9659090638160706]
```

Random Forest Classifier

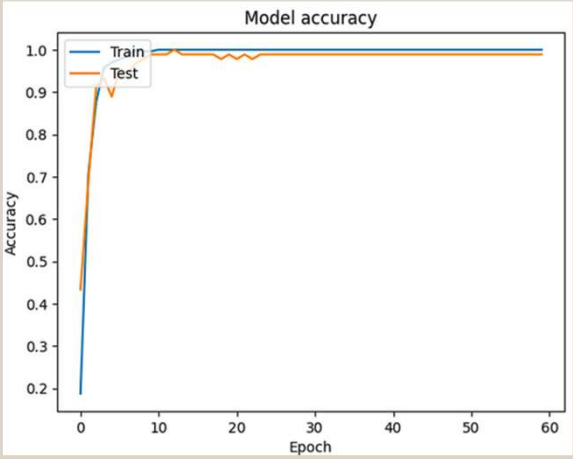


Classifiers Accuracy Comparison

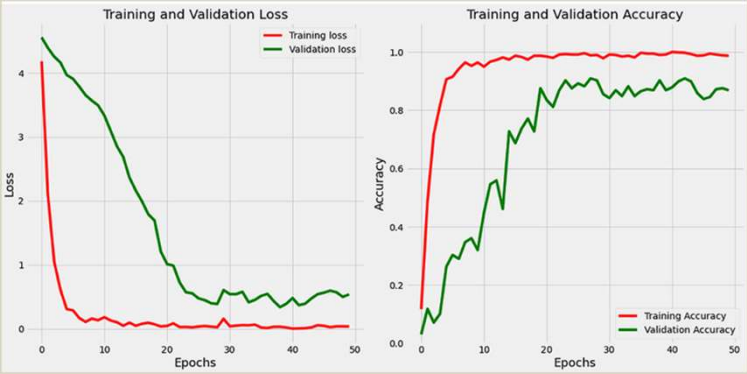


Leaf CLassification

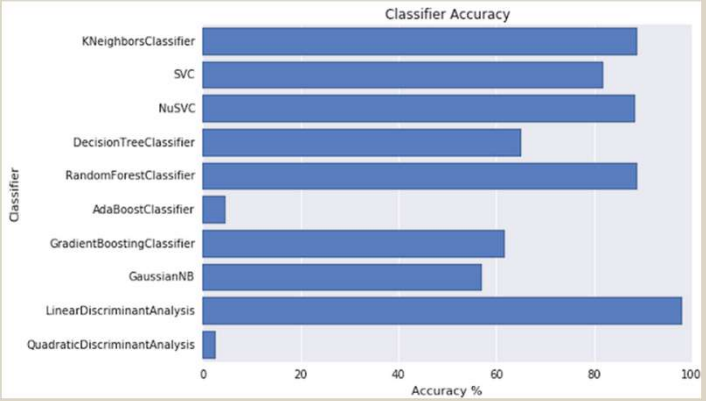
LDA Classifier



ResMLP

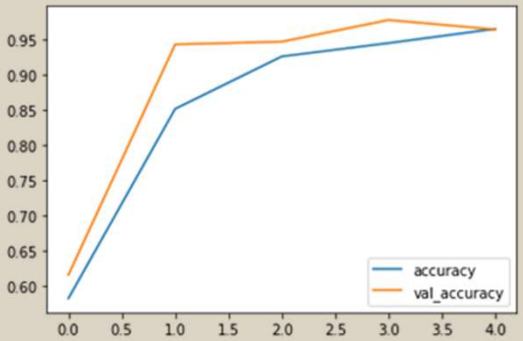


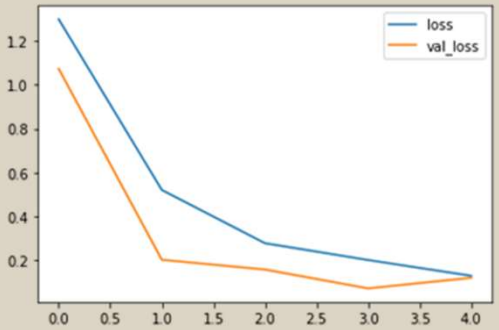
Machine Learning Classifiers



Paddy Disease

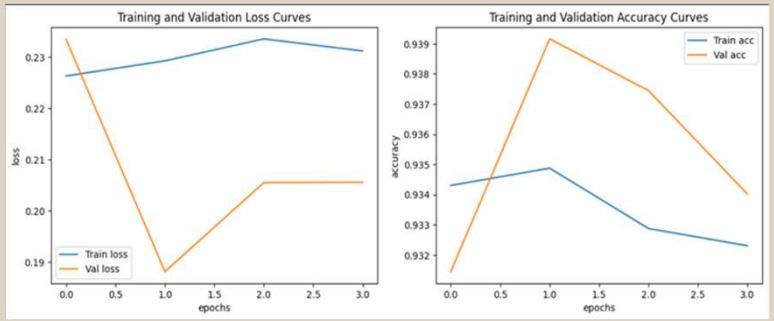
ResNet34 with K_fold



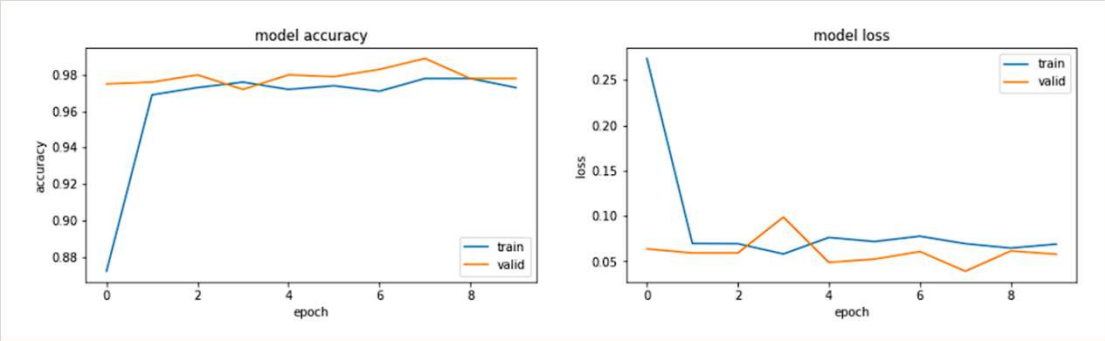


VGG 19

Cotton weeds



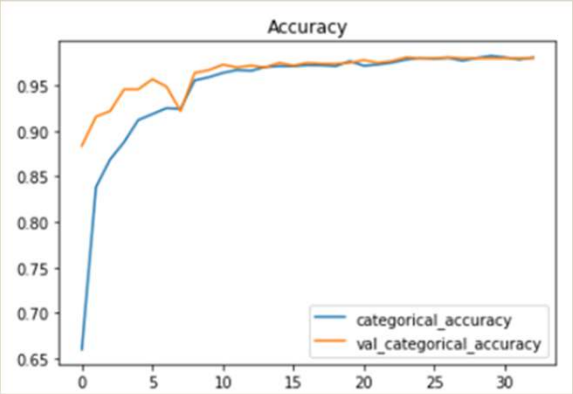
ResNet50 with K_fold

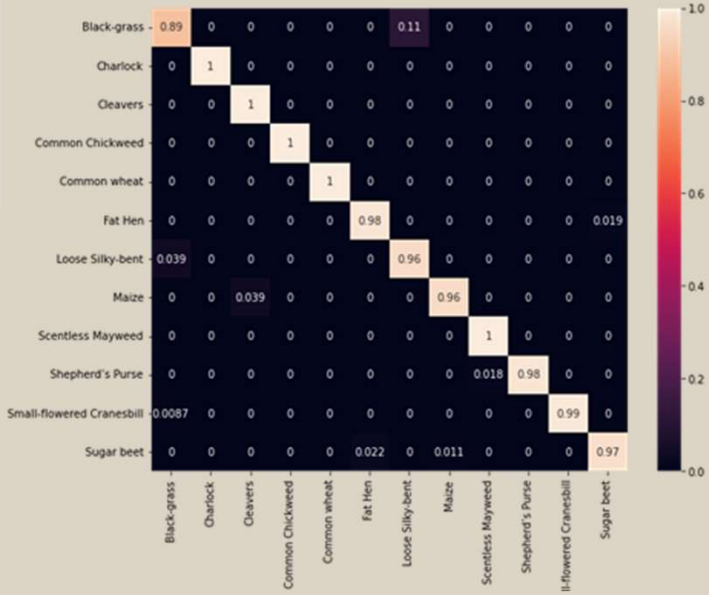


	fold	train loss	validation loss	train accuracy	validation accuracy
0	1	0.226351	0.233549	0.934304	0.931448
1	2	0.229307	0.188105	0.934876	0.939160
2	3	0.233598	0.205504	0.932876	0.937446
3	4	0.231248	0.205573	0.932305	0.934019

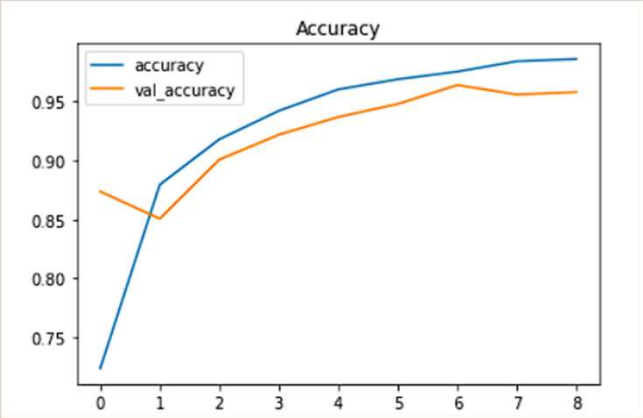
Seedlings

EfficientNetB2



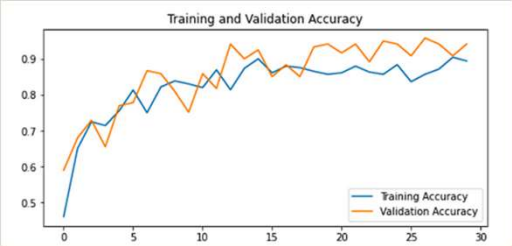


MobileNetV2



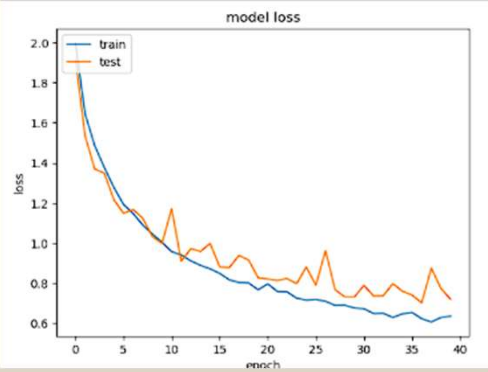
Flower Classification

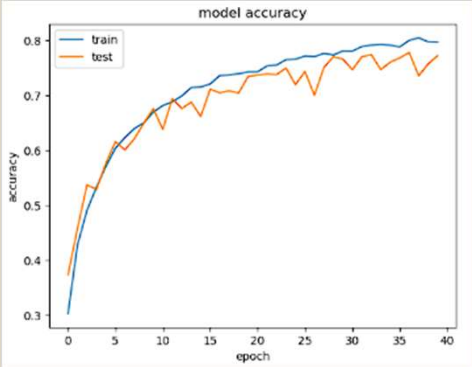
Pytorch Resnet50 KFOLD



	fold	train loss	validation loss	train accuracy	validation accuracy
0	1	0.299706	0.245819	0.907474	0.925748
1	2	0.280097	0.236445	0.913404	0.928007
2	3	0.292233	0.275498	0.905318	0.916973
3	4	0.281100	0.295912	0.912376	0.907371

CNN





FUTURE WORK

IoT system implementation

Improve Accuracy

Drone

Continue App Development

GRANTS & ACHIEVEMENTS

1st Place Climathon EUI 2022 Hackathon Winners (Empowering Theme)

The Academy of Scientific Research & Technology Fund

Dell Technologies Graduation Project Competition Finalists (Top 27 Regional)

_VOIS Shortlisted VOIS Hackathon (Vodafone) National Competition

YLF YOUTH LEADERS FOUNDATION مؤسسة شباب القادة Youth leaders' foundation (YLF) Climate Leaders Finalists

وزارة التعليم العالي والبحث العلمي MINISTRY OF HIGHER EDUCATION AND SCIENTIFIC RESEARCH Hack4climate "Create to Sustain" Hackathon Finalists

THANK YOU

Together

Business Model Canvas

01 Key Partners

- Agricultural technology providers
- Manufacturers and suppliers of eco-friendly products
- Local governments and regulatory bodies
- Environmental organizations
- Influencers
- Ministry of Agriculture and Land Reclamation

02 Key Activities

- Development of AI models
- App & Platform development
- Marketing and promotion
- Development of IoT system
- Sales and partnerships

03 Key Resources

- IoT & hardware components
- Partnerships
- AI/cloud services
- ML engineers mobile developers

04 Value Proportions

- Community for environment enthusiasts
- All-in-one climate change app
- Increase awareness levels in a trendy way
- Affordable & easy planting smart guide
- Climate-positive Farming System for increasing crop productivity
- Providing online eco-friendly products and gardening supplies

05 Customer Relationships

- Collaborative and supportive community
- Interactive and informative user experience
- Professional and responsive relationships with eco-friendly product providers

06 Channels

- Online marketplaces for the eco-friendly product marketplace
- Mobile app
- Social media
- Agricultural landowners and companies through events & advertisements

07 Customer Segments

- Climate change enthusiasts
- People looking for eco-friendly products
- Environmental enthusiasts
- Home farming and gardening enthusiasts
- Eco-friendly product manufacturers
- Agricultural landowners and companies

08 Cost Structure

- Cloud services
- Mobile app development
- Partnership and collaboration costs with key partners
- Marketing & advertising
- IoT system infrastructure

09 Revenue Streams

- Ads
- Subscriptions for premium services
- Sales of agricultural services as a package
- Commission on eco-friendly products

Market Analysis

Competitive analysis

After conducting a comprehensive market research on Climate change apps, we found that most apps focus on tracking Co2 and providing tips to reduce it, other apps offer scientific visualizations and climate change data but this will be only helpful for tech savvy, we've also considered other indirect competitors like planting apps (planta, picture this and smart plant home) and event organizing apps (Meetup and Eventbrite) in addition to community platforms like (Quora and Reddit), below is a cutdown version of our competitive analysis.

	Changeit	Earth hero	We don't have time	Carbon Footprint and Co2 tracker	Smart Plant home	Planta	Meetup	Together
Actions to reduce CO2	✓	✓	✗	✓	✗	✗	✗	✓
Points/awards	✗	✓	✗	✗	✗	✗	✗	✓
Pollution reporting	✗	✗	✗	✗	✗	✗	✗	✓
Articles, News, Engagement	✗	✗	✓	✓	✗	✗	✗	✓
Planting smart guide	✗	✗	✗	✗	✓	✓	✗	✓
Event organization	✗	✗	✗	✗	✗	✗	✓	✓

EXCUTIVE SUMMARY

Together is a mobile application that will help in raising awareness and understanding of how simple human actions impact climate change, in addition to engaging, and building a growing community of people concerned with climate change, by focusing on habit building, thus empowering communities to increase resilience against climate change.

Planta (top indirect competitor)

2 million users
2nd top grossing on playstore

Many people care about plants but find it hard to take good care of them, **Planta provides them with the help they need.**

In our app "Together" we provide this through the planting smart guide, and it's customized to Egypt's weather conditions for better guidance.